

Cyber Security and Safety – Evolving Fast

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What do we need to talk about?

- No safety without secure systems and Opsec processes
- Protect against the “abuse” case at every level of tech
- Examples from case studies over 27 years
- OT/PCS security is different from IT security
- Cyber security’s strategy is not safe
- Pen testing != assurance
- Action Plan – Hope is not a strategy

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Jaguar Land Rover posts heavy loss after cyber-attack

14 November 2025

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Theo Leggett

Business correspondent



Triton: hackers take out safety systems in 'watershed' attack on energy plant

Sophisticated malware halts operations at power station in unprecedented attack which experts believe was state-sponsored



Davis-Besse Nuclear Plant Infection

2003 • 2000-2009

Target

Davis-Besse Nuclear Power Plant

Hackers heat French supermarket fridges, sparking fires and spoiling food

By **Leo Marchandon** and **Gabriel Stargardter**

October 12, 2026 11:57 PM GMT+10 · Updated October 14, 2026

Aa



Maroochy Water Services Attack

Water

2000 • 2000-2009

Target

Maroochy Shire Council

Method

Insider Attack, Radio Transmitter

Immediate Impact

Release of 800,000 liters of raw sewage into local parks, rivers and a hotel property.

- 2024 – Albert River
- 450 million L – ***burst pipe***

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It’s about Abuse

- Misusing features
- Exploiting trust
  - e.g. IP addresses,
  - clear text networks
  - Injection attacks
  - Downloads
- Security leads to loss of access
- Convenience – good for users & good for malware and attacks

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Common Issues

- Mean time to failure assumes benign threats
- Safety shares networks with process control
- Controls weak in the face of cyber threats
  - No authorised patches from control vendors
  - Patching does not remove all the bugs
    - Reduces probability though . . .
  - Common and default passwords
  - No encryption on radio, etc

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IT’s Cyber Plan = Caution - 1

- Internet was not design to be secure

- Patch, patch, patch
  - But AI will just find more bugs
  - Time to recertify
- Default deny
- Vendor generic accounts
- Professional versus – learn on the job?

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IT’s Cyber Plan = Caution - 2

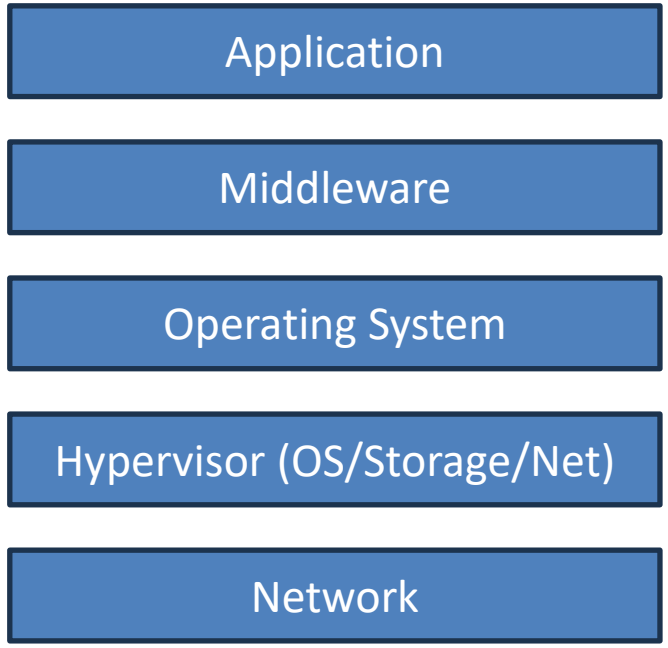
- Antivirus and more complex IT tools on the OT devices
- More connectivity for tools
- Virtualisation and cloud
- Low security defaults

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Security and the “IT” stack

- Secure config
- Accounts and identity
- Vulnerability mitigation
- Secure access (remote?)
- Secure comms
- Logs and monitoring



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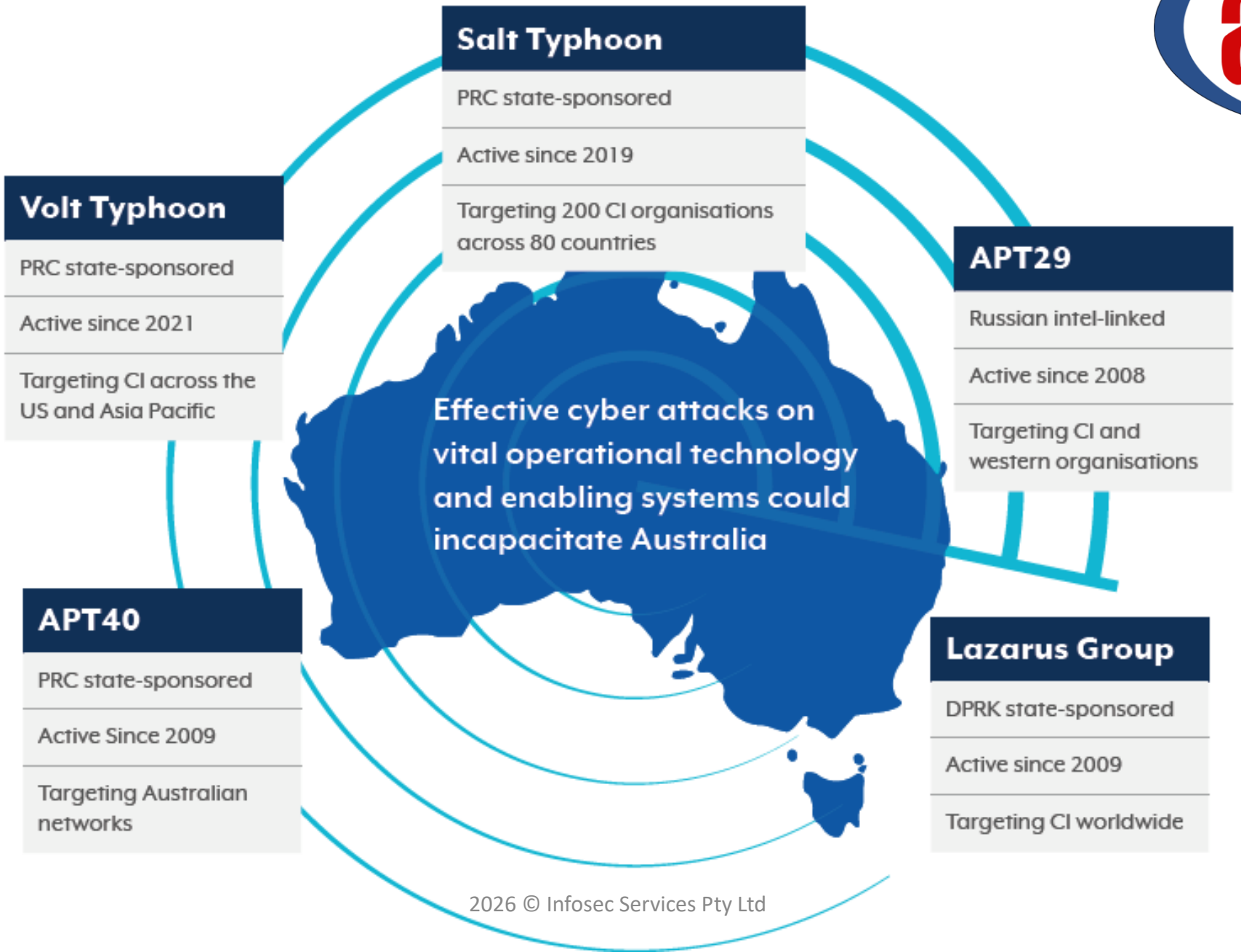
Cyber Threat Landscape

- Range of attackers – not just kids
- Increasing focus on Critical Infrastructure (extortion etc)
- Advanced Persistent Threats
  - Hack now, exploit later at the worst time
- Rockwell –
  - 3% - loss of safety systems
  - 84% - enter via IT

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State-sponsored actors are targeting Australia



ACSC

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Pen Testing != Assurance

- Risky – How will controllers react?

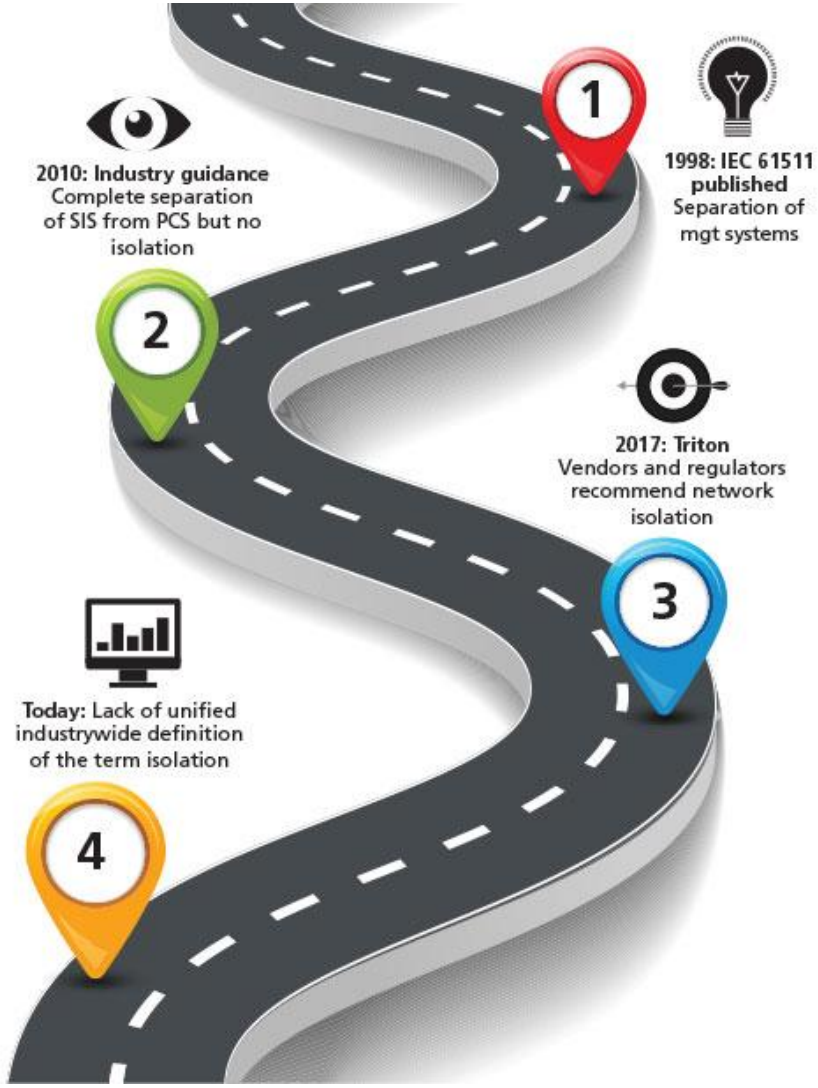
- ASAE 3000:
  - sufficient appropriate evidence . . . . degree of confidence . . . . the outcome of the measurement or evaluation of an underlying subject matter against criteria
- Pen testing – not cover the full range of controls nor compensating controls → misleading results
- Reports often confuse vulnerability severity with risk

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Action Plan - 1

- Cyber Awareness for
  - safety engineers
  - OT operators
- Collaboration
- Separate security for safety systems
  - No reuse - passwords, networks etc
- Independent safety control systems
- One way interconnections with corporate
  - Data diodes or at least
  - No incoming files, read only access



ISA Diagram

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Action Plan - 2

- IEC 62443 Security Program, IEC 61511
- CI Fortify, ASD Essentials for OT
- Resilience design & sound implementation
- Threat Assessment (vis Standards Australia Handbook 167)
- Govt – Cyber Industry Professionalisation – ACS lead
- Security planning based on SIL

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ASD Essentials for OT

- ACSC OT Principals – Safety is Paramount
- Notes Safety as key goal
  - Not separate requirements for safety systems
- Priorities:
  - Critical processes
  - Boundary Services
- Based on Info Security Manual – and incidents
- Exercise running isolated OT

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TIME FOR COFFEE ?

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